Water Pollution

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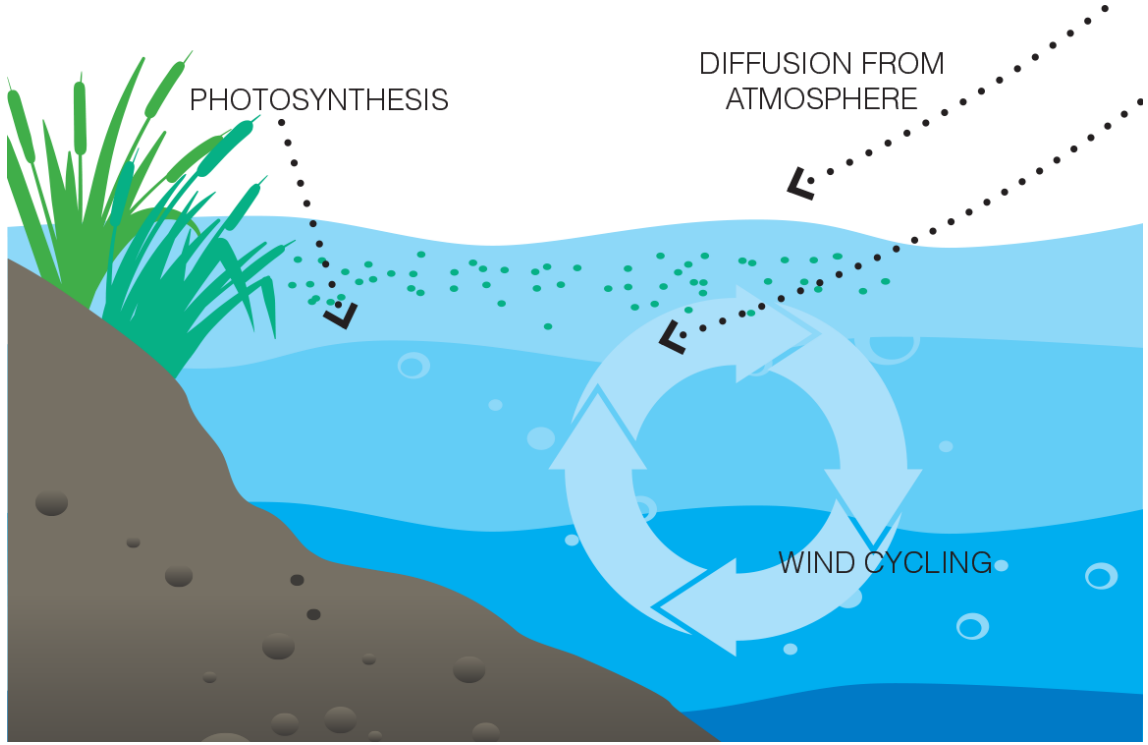
Dissolved Oxygen

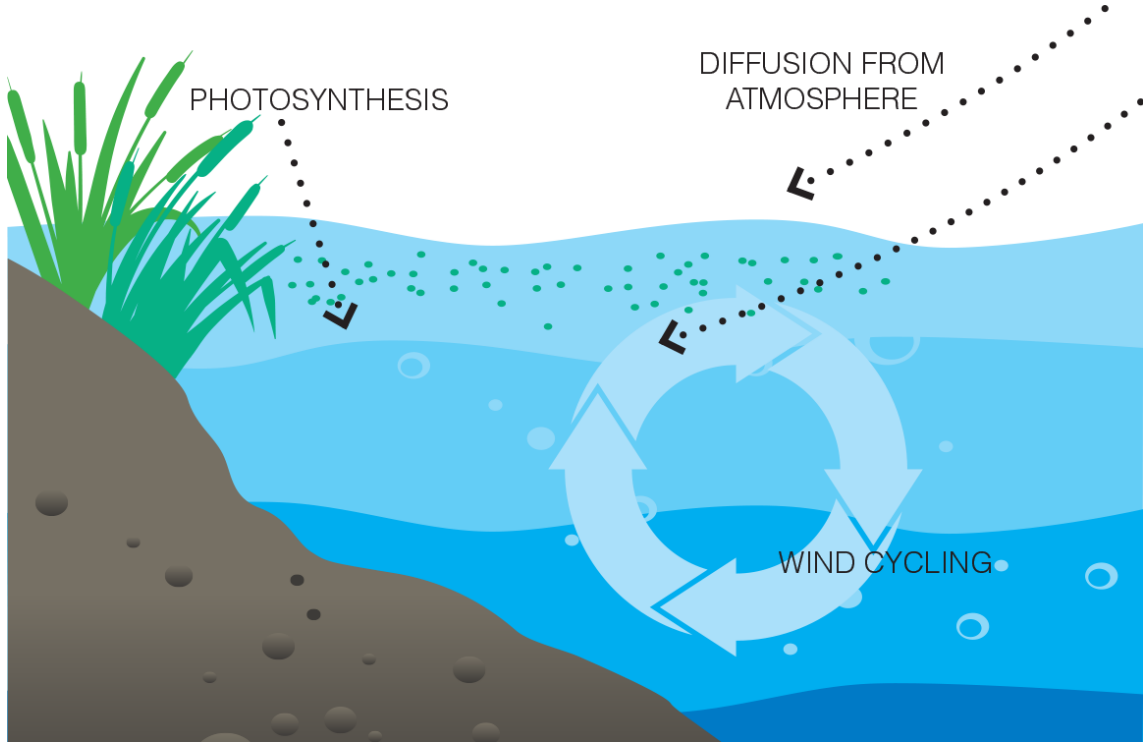
- Dissolved oxygen (DO) is the amount of oxygen that is present in water.
- Water bodies receive oxygen from the atmosphere and from aquatic plants.
- Running water, such as that of a swift moving stream, dissolves more oxygen than the still water of a pond or lake.
- All aquatic animals need DO to breathe. Low levels of oxygen (hypoxia) or no oxygen levels (anoxia) can occur when excess organic materials, such as large algal blooms, are decomposed by microorganisms.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Dissolved Oxygen, EPA, https://www.epa.gov/national-aquatic-resource-surveys/indicators-dissolved-oxygen#:~:text=What%20is%20dissolved%20oxygen%3F,of%20a%20pond%20or%20lake. 3. Image: https://www.horiba.com/int/water-quality/applications/agriculture-crop-science/dissolved-oxygen-in-greenhouse/

How does oxygen get into water?

- Oxygen from the atmosphere dissolves and mixes into the water's surface.
- Algae and underwater grasses release oxygen during photosynthesis.
- Ocean waters generally have more oxygen. River waters are fast-moving, which helps oxygen from the air mix in.



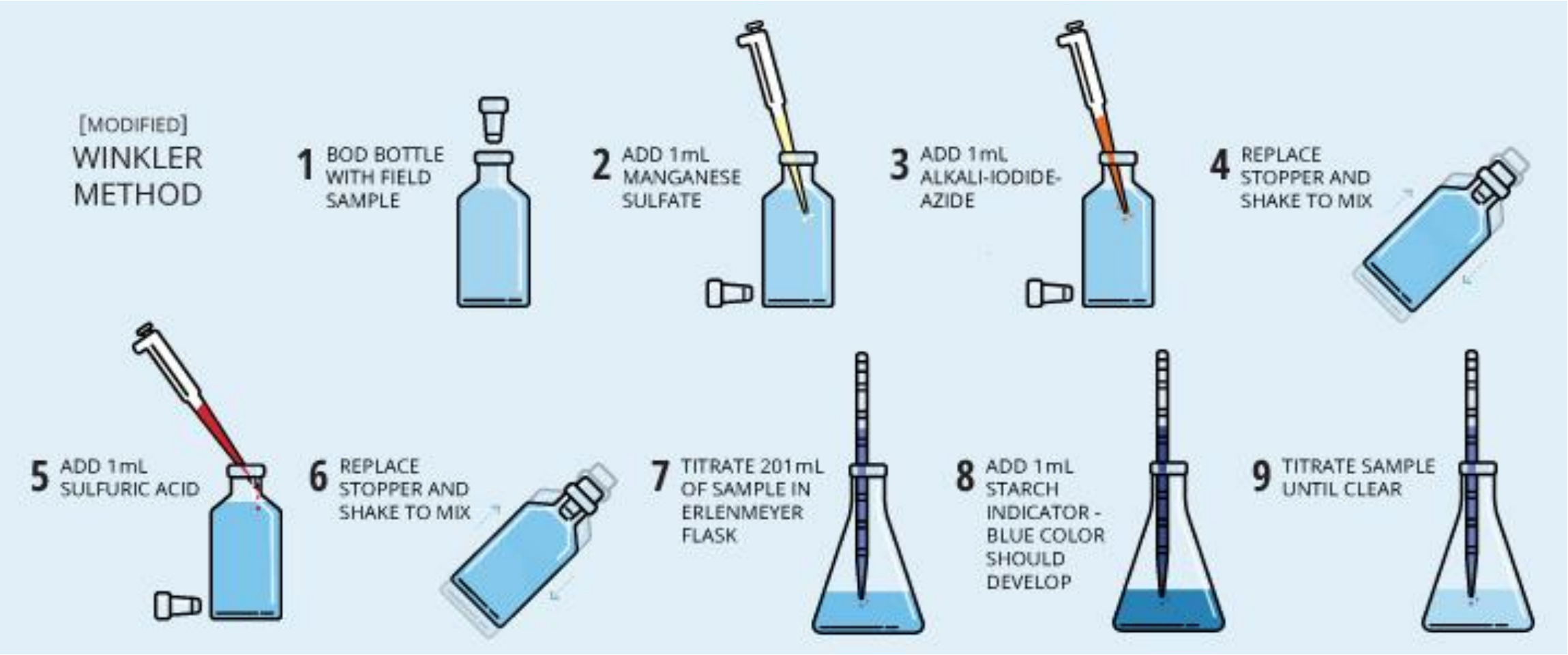
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 3. Image: https://www.eutechinst.com/pdt-para-do-eutechdo450.html

Dissolved Oxygen

- For living organism, about 4 mg/L of minimum DO should be in water.
- Oxygen is considered as poorly soluble in water.
- Its solubility is related to pressure and temperature.
- In freshwater, DO reaches 14.6 mg/L at 0 °C and approximately 9.1, 8.3, and 7.0 mg/L at 20, 25, and 35 °C, respectively, at 1 atm pressure.
- At temperatures of 20 and 30 °C, the level of saturated DO is 9.0-7.0 mg/L.
- Detemination by **WINKLER'S METHOD.**

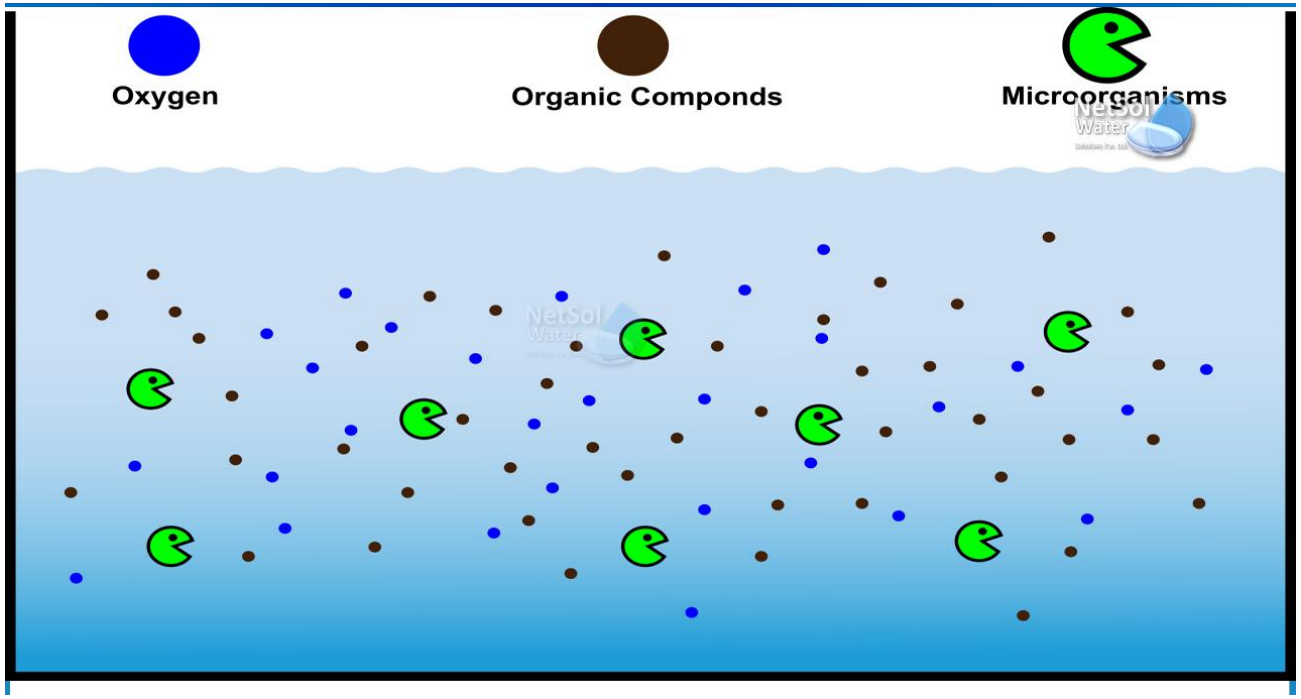
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.sciencedirect.com/science/article/abs/pii/B9780128023266000022

Determination of DO



Biochemical Oxygen Demand (BOD)

- Biodegradable materials consists of organics that can be utilized for food by naturally occurring microorganisms within a reasonable length of time.
- Biochemical oxygen demand (BOD) represents the amount of oxygen consumed by bacteria and other microorganisms while they decompose organic matter under aerobic (oxygen is present) conditions at a specified temperature.
- Intention: To measure the strength of organic matter in wastewater



$$C_XH_YO_ZN\ (\text{Organics}) + O_2 \xrightarrow{\textit{Microorganisms}} CO_2 + H_2O + \text{New Cells}$$

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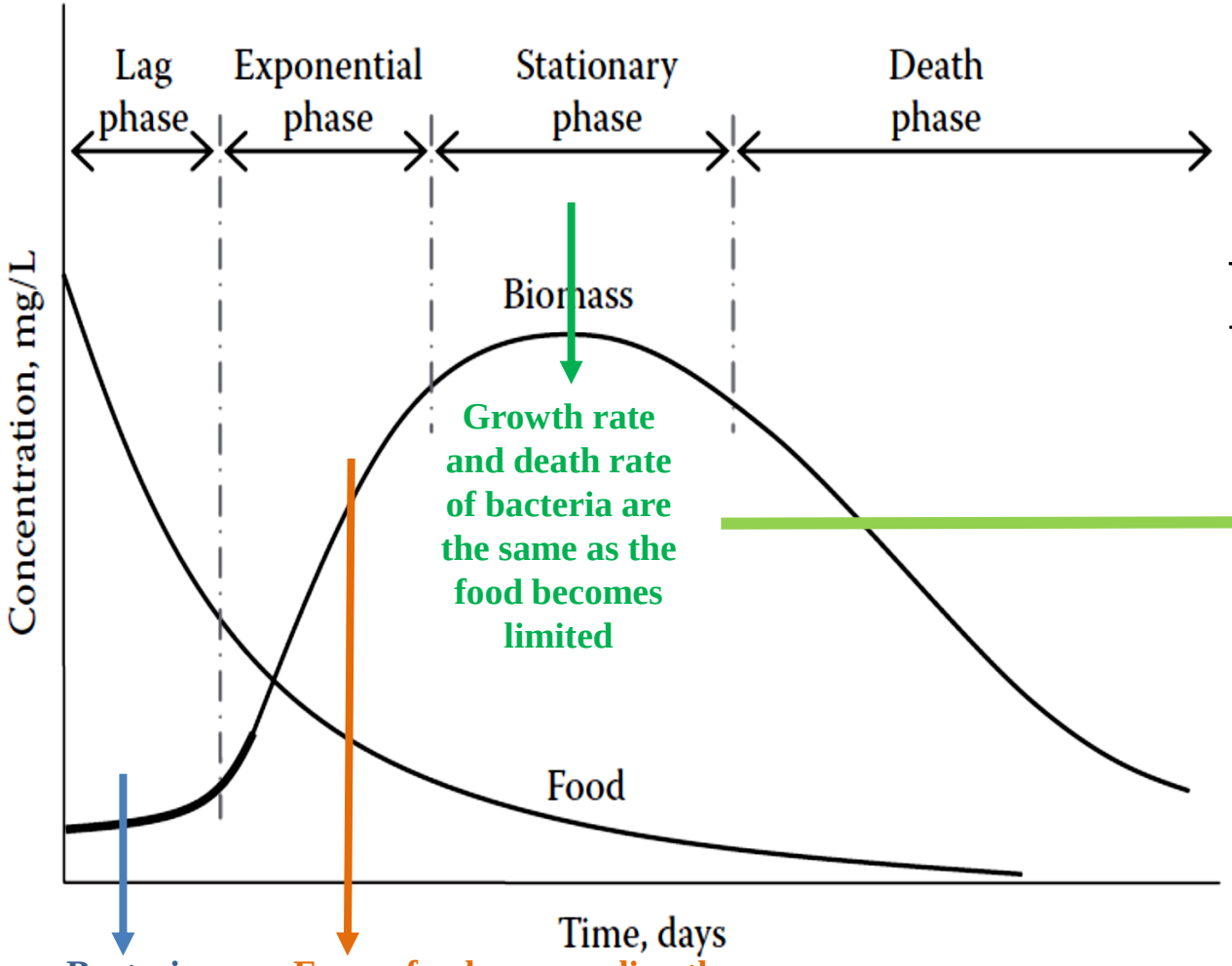
Growth Curve of Microorganisms



Growth rate and death rate of bacteria are the same as the food becomes limited

Bacteria acclimate to the new environment

Excess food surrounding the bacteria; rate of metabolism and growth is a function of the ability of microorganism to process the substrate

The rate at which organics are utilized by micro organisms is a first-order reaction-

$$\frac{dX}{dt} = kX$$

Where,

– $\frac{dX}{dt}$ is the growth rate of the biomass (mass dry wt/vol.t)

–X is cell concentration (mass dry wt/vol)

– k is the specific growth rate (time-1)

– t is time

Numerical

E. coli cells take 2 hrs. produce 1024 cells. How much time they will take to produce 2048 cells? (Generation time of *E.coli* is 20 min)

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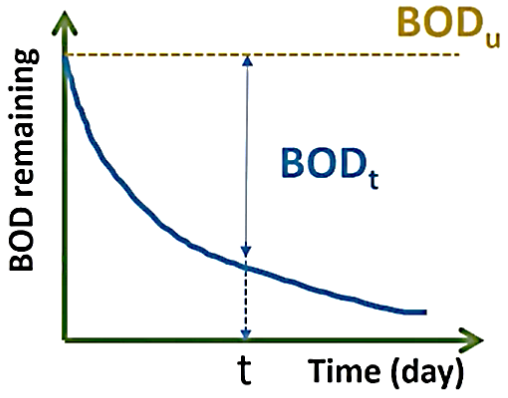
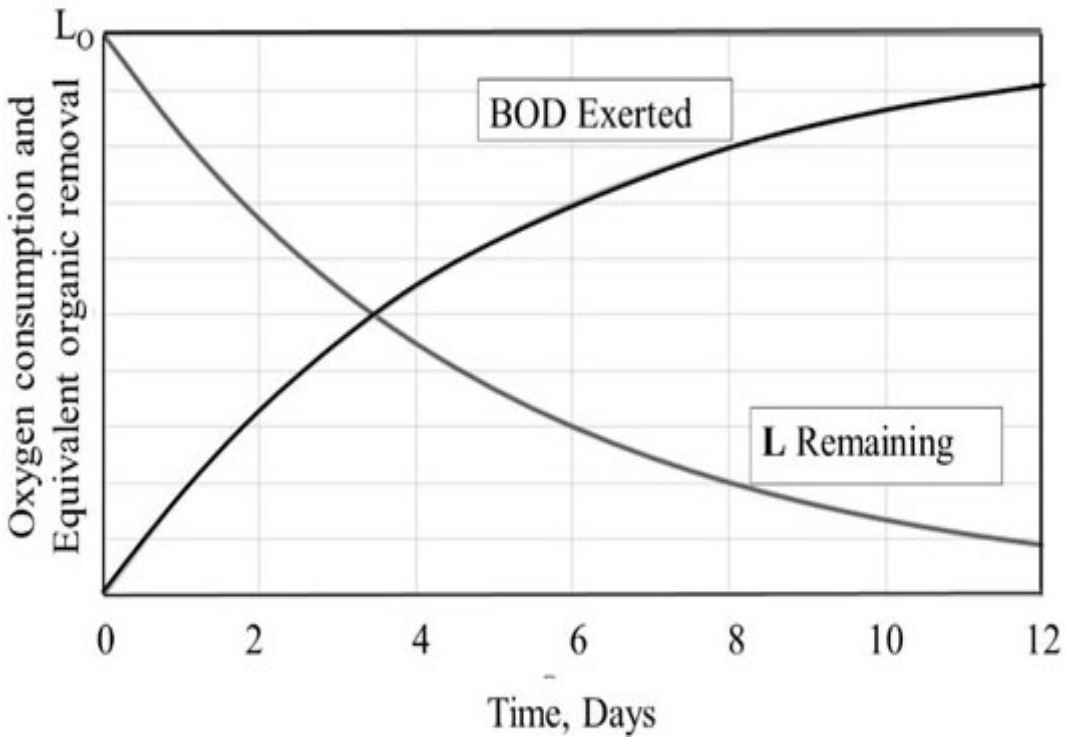


BOD$_u$ (ultimate BOD)

Amount of oxygen needed to decompose all the organic matter (mg O_2/L)

BOD$_t$

Oxygen consumed in water after t days (mg O_2/L)



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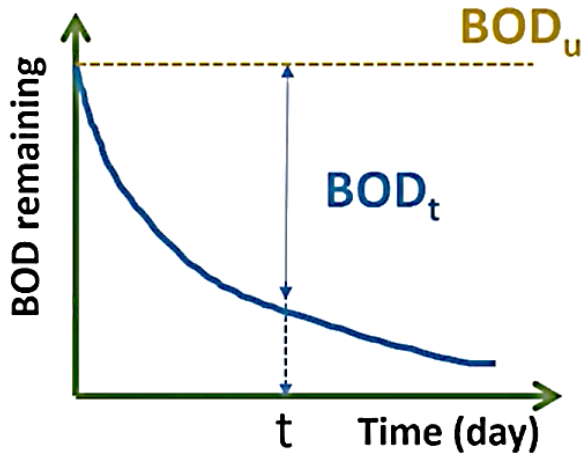

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BOD Measurement

For BOD, Saturation oxygen concentration of 9 mg/L in dilution water, with a minimum of 2 and a maximum of 7 mg/L of O_2 being consumed.

Titrant: Sodium Thiosulfate

BOD_5 is calculated by:

- Unseeded : $\mathrm{BOD}_5 = \dfrac{(D_0 - D_5)}{P}$
- Seeded: $\mathrm{BOD}_5 = \dfrac{(D_0 - D_5) - (B_0 - B_5)f}{P}$

where:

D_0 is the dissolved oxygen (DO) of the diluted solution after preparation (mg/L)

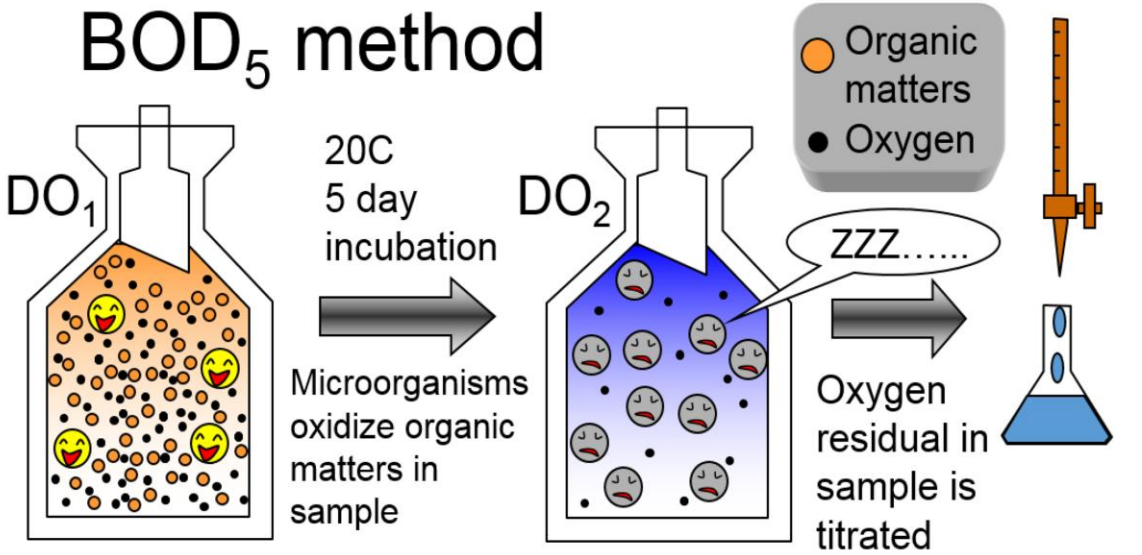
D_5 is the DO of the diluted solution after 5 day incubation (mg/L)

P is the decimal dilution factor

B_0 is the DO of diluted seed sample after preparation (mg/L)

B_5 is the DO of diluted seed sample after 5 day incubation (mg/L)

f is the ratio of seed volume in dilution solution to seed volume in BOD test on seed



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Numerical

The BOD of a wastewater is suspected to range from 50 to 200 mg/L. Three dilutions are prepared to cover this range. The procedure is the same in each case. First the sample is placed in the standard BOD bottle and is then diluted to 300 mL with organic-free, oxygen-saturated water. The initial dissolved oxygen is determined and the bottles tightly stoppered and placed in the incubator at 20°C for 5 days, after which the dissolved oxygen is again determined. Determine the BOD of the sample.

Wastewater, milliliter	DO_i, mg/L	DO_5, mg/L
5	9.2	6.9
10	9.1	4.4
20	8.9	1.5

Thank you

For any questions, please feel free to reach out to me.